MSc Data Science

DS601: Mathematical Foundations for Data Science **Date:07/09/2026**

Question 1 — Random Variable

Solution

(a) Let X be the number of heads obtained in three fair-coin tosses. Hence,

$$X \in \{0, 1, 2, 3\}.$$

(b) Since X ~ Binomial(n = 3, p = 1/2),

$$P(X = x) = C(3,x)(1/2)^{x}(1/2)^{3-x} = C(3,x)/8.$$

x	0	1	2	3
P(X=x)	1/8	3/8	3/8	1/8

(c) $P(X \geq 2) = P(X=2) + P(X=3) = 3/8 + 1/8 = 1/2$.

Answer: P(X ≥ 2) = 0.5.

(d) $E[X] = np = 3(1/2) = 3/2$.

Answer: E[X] = 1.5.

Question 2 — Probability Distribution

Solution

Given P(X=x) = k(x+1), for x = 0,1,2,3,4.

Using ΣP(X=x)=1:

$$k[(1)+(2)+(3)+(4)+(5)] = 1 \Rightarrow 15k = 1 \Rightarrow k = 1/15.$$

x	0	1	2	3	4
P(X=x)	1/15	2/15	3/15	4/15	5/15

(a) $P(1 < X \leq 3) = P(X=2)+P(X=3) = 3/15+4/15 = 7/15$.

(b) $P(X \geq 2) = (3+4+5)/15 = 12/15 = 4/5$.

(c) $E[X] = \Sigma x P(X=x) = (0\cdot1 + 1\cdot2 + 2\cdot3 + 3\cdot4 + 4\cdot5)/15 = 40/15 = 8/3$.

(d) First, $E[X^2] = (0^2\cdot1 + 1^2\cdot2 + 2^2\cdot3 + 3^2\cdot4 + 4^2\cdot5)/15 = 130/15 = 26/3$.

$$Var(X) = E[X^2] - (E[X])^2 = 26/3 - (8/3)^2 = 14/9.$$

Answers: 7/15, 4/5, 8/3, and Var(X) = 14/9.

Question 3 — PMF, PDF and CDF

(A) Continuous Random Variable

Solution

For X uniformly distributed over [0,10]:

(a) The PDF is

$$f_X(x) = 1/10,\ 0 \le x \le 10;\ 0 \text{ otherwise.}$$

(b) The CDF is

$$F_X(x) = 0 \text{ for } x<0; \quad x/10 \text{ for } 0\le x\le 10; \quad 1 \text{ for } x>10.$$

(c) $P(2 < X < 7) = (7-2)/(10-0) = 5/10 = 1/2$.

(d) Using the CDF:

$$P(2 < X < 7) = F(7) - F(2) = 7/10 - 2/10 = 1/2.$$

Answer: P(2 < X < 7) = 0.5.

Question 4 — Binomial Distribution

Solution

(a) $X \sim \text{Binomial}(n = 12, p = 0.75)$.

$$P(X=k) = C(12,k)(0.75)^k(0.25)^{12-k}.$$

(b) $P(X=9) = C(12,9)(0.75)^9(0.25)^3 \approx 0.258104$.

(c) $P(X \ge 8) = \Sigma$ from k=8 to 12 $P(X=k) \approx 0.842356$.

(d) By conditional probability,

$$P(X=10 \mid X\ge 8) = P(X=10)/P(X\ge 8) \approx 0.232293/0.842356 \approx 0.275766.$$

(e) We need the smallest integer r for which $P(X \ge r) < 0.10$.

$P(X \ge 11) \approx 0.158382 > 0.10$, while $P(X \ge 12) = (0.75)^{12} \approx 0.031676 < 0.10$.

Answer: r = 12.

Question 5 — Uniform Distribution

Solution

$X \sim \text{Uniform}(800,1200)$, so the interval length is $1200-800 = 400$ hours.

(a) $P(X > 1050) = (1200-1050)/400 = 150/400 = 0.375$.

(b) $P(900 < X < 1100) = (1100-900)/400 = 200/400 = 0.5$.

(c) “At least another 200 hours” after 900 hours means $X \ge 1100$. Therefore,

$$P(X \ge 1100 \mid X > 900) = P(X \ge 1100)/P(X > 900) = (100/400)/(300/400) = 1/3.$$

(d) $P(X < a) = (a-800)/400 = 0.35$.

$$a - 800 = 140 \Rightarrow a = 940 \text{ hours.}$$

Answers: 0.375, 0.5, 1/3, and a = 940 hours.

Question 6 — Function of a Random Variable

Solution

Given $f_X(x) = 2x$ for $0<x<1$, and $Y = X^2$.

Since $0<X<1$, we also have $0<Y<1$.

(a) For $0<y<1$,

$$F_Y(y) = P(Y \le y) = P(X^2 \le y) = P(X \le \sqrt{y}) = \int_0^{\sqrt{y}} 2x\,dx = y.$$

Thus $F_Y(y) = 0$ for $y \le 0$; y for $0<y<1$; 1 for $y \ge 1$.

(b) Differentiating the CDF:

$$f_Y(y) = 1 \text{ for } 0<y<1, \text{ and } 0 \text{ otherwise.}$$

(c) $P(1/4 < Y < 3/4) = F_Y(3/4) - F_Y(1/4) = 3/4 - 1/4 = 1/2$.

(d) $E[Y] = \int_0^1 y\, f_Y(y)dy = \int_0^1 y\,dy = 1/2$.

Answers: Y ~ Uniform(0,1), P(1/4<Y<3/4)=1/2, and E[Y]=1/2.